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Marketing the Sustainability Initiatives in India: The Status of Green House Gas (GHG) Emissions and the Carbon Market

Tinu Joseph

Abstract

This study examines domestic and international efforts to market sustainability initiatives by examining the state of GHG emissions and the development of the carbon market, with a particular emphasis on India. International accords, starting with the 1972 Stockholm Conference and ending with the legally binding emission reduction targets set by the Kyoto Protocol (1997), were prompted by global concerns over resource overuse and ensuing climate shifts. The Emissions Trading System (ETS), Clean Development Mechanism (CDM), and Joint Implementation (JI) are the three crucial market-based mechanisms that were developed by the Protocol to solve environmental market failure by internalizing the societal cost of pollution. By creating Nationally Determined Contributions (NDCs) and mandating that all parties, especially developing countries, commit to aggressive carbon reduction targets, the ensuing 2015 Paris Agreement solidified this direction. An analysis of India's industrial landscape's Global Warming Potentials (GWP) between 2005 and 2018 shows alarming patterns, including an almost universal increase across industries. Importantly, GWP increases of more than 268% were recorded in sectors like Non-Ferrous Metals, Other Energy Industries, and Chemicals & Fertilizers. With 36 nations implementing ETS and 39 enacting national carbon taxes, the use of carbon pricing schemes has increased globally. But taken as a whole, these tools only account for 24% of world emissions. Although the European Union ETS is the largest compliance market in the world, major compliance markets such as China and New Zealand saw substantial price fluctuation in the early months of 2023, which raised concerns about the stability and effectiveness of the market. By tokenizing carbon credits and turning them into digital assets on block chain systems, as demonstrated by exchanges like Toucan Protocol, the industry is also seeing technological advancement. India has introduced the Carbon Credit Trading Scheme (CCTS), 2023 in accordance with its NDC pledge to reduce carbon emissions by 45% by 2030 (compared to 2005 levels) and achieve net-zero emissions by 2070. The Grid Controller of India (GCI) and the Central Electrical Regulatory Commission (CERC) oversee registry and trade operations, respectively, while the Bureau of Energy Efficiency (BEE) administers the CCTS. National carbon neutrality programs that concentrate on afforestation, clean air projects, and the creation of Carbon Capture Utilization and Storage (CCUS) frameworks to alleviate the industry's excessive reliance on coal are employed in conjunction with these efforts. The paper advocates for a community-involved, government-regulated system for the payment and control of CO₂ emitters, arguing that a more comprehensive and effective regulation model may need to go beyond merely market-oriented measures in order to assure true sustainability.

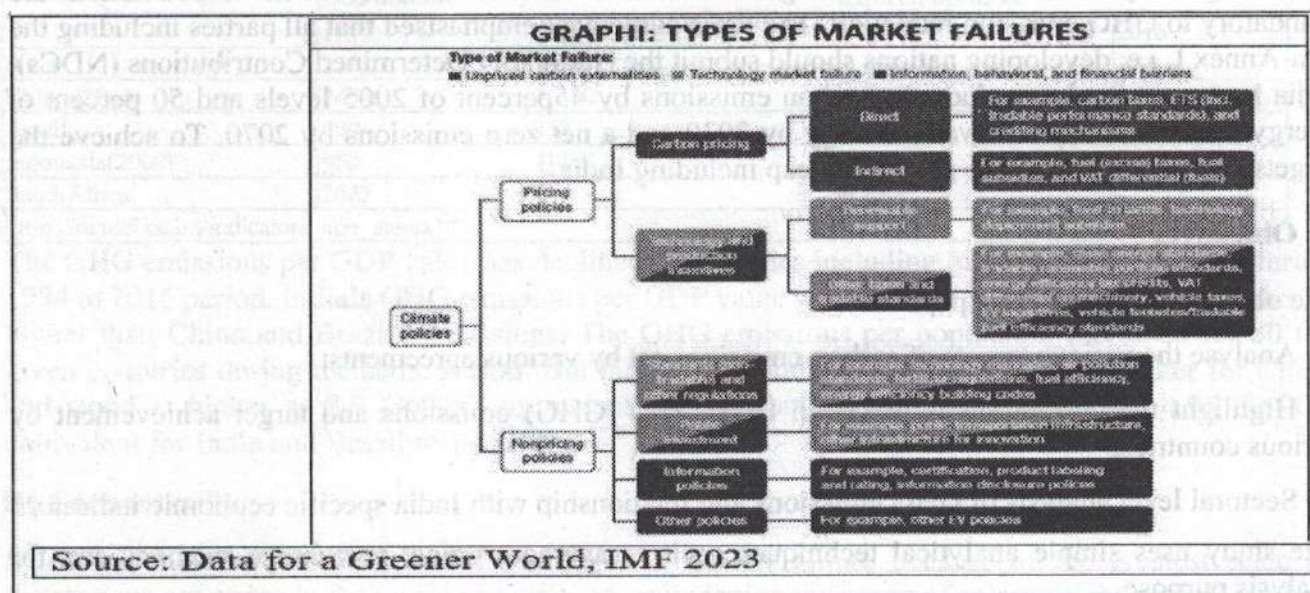
Keywords: Sustainability, Green House Gas, Carbon Tax, Carbon Credit, Climate Change

Introduction

The ongoing exploitation of earth's resources and its overutilization in name of 'development' led to environmental damages and consequently harsh climatic changes. The world nations under the United Nations has come up with policies to curtail the over utilisation of resources by putting targets to reduce the carbon emissions and marketize the green initiatives.

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The United Nations Conference on Human Environment usually referred as 'Stockholm Conference' in 1972 led to the starting point of need for international environment policies. The United Nations Environmental Programme (UNEP) was established in 1972 to address the environmental challenges and state cooperation in this matter. The concept of carbon trading may have its theoretical background on market failures in environmental pollution due to negative externalities and not reflecting the social cost in actual price of products. Graph I addresses the types of market failures.



The climate change policies include pricing and non-pricing policies under the principles of Paris agreement. The Carbon pricing was part of pricing policies adapted for mitigating the GHG emissions under climate change process. According to records the first tradable emission mechanism was the United States Acid Rain programme under the US Clean Air Act Amendment, 1990. It was a Cap-Trade programme for sulfur dioxide allowing companies to buy and sell emission allowances thereby creating a market for pollution reduction. Another landmark in this area is the United Nations Conference on Environmental Development (UNCED) via 'Earth Summit' in 1992 came out with 'Rio declaration' with 21 agendas and United Nations Framework Convention on Climate Change (UNFCCC) for environmental protection and sustainable development. Later in 1997 the Kyoto Protocol adopted in Japan as an extended arm of UNFCCC with a through commitment from the member states to reduce the Green House Gas (GHS) emissions. This protocol came into force on February 16, 2005 with a period wise commitment. The first period of 2008 to 2012 was committed to reduce GHG emissions by an average of 5 percent below the 1990 levels. The Kyoto Protocol set up the base for the establishment of carbon markets. The emission reductions were quantified and can be traded as carbon credits according to Article 17 of Kyoto Protocol.

The Kyoto protocol introduced market-based mechanisms called Emissions Trading System (ETS),

Clean Development Mechanism (CDM) and Joint Implementation (JI) for carbon trading. The 'Doha amendment' came up with extended Kyoto Protocol's commitment period towards 2012 to 2020. The emission reduction targets were updated to 18 percent below 1990 levels. This agreement extended the commitment reduction period of developed nations to 2020 and nitrogen trifluoride was also added to the list of GHGs. The Annex I party lists were prepared via Copenhagen Accord and Cancun agreements which mostly consists rich countries and non-Annex I parties consists of developing and other countries. The Paris agreement adopted in 2015 set the limit of global temperature increases to

below 2 degrees Celsius above pre-industrial levels and urged nations to limit the increases to 1.5 degree Celsius (UNFCCC). It set nationally determined contributions (NDCs) in which adopt actions for GHG reductions, formulating long-term strategies, financial aid and capacity building along with technology transfer for reduction of GHGs. The global market for carbon credits, the Paris Agreement Crediting Mechanism (PACM) overseen by supervisory body responsible to parties to Paris agreement.

According to Kyoto Protocol and Paris Agreement annex – I countries, i.e. the developed nations are mandatory to GHG emission reduction. The Paris agreement emphasised that all parties including the non-Annex I, i.e. developing nations should submit the Nationally Determined Contributions (NDCs). India has committed to reduce the carbon emissions by 45 percent of 2005 levels and 50 percent of energy needs meet by renewable energy by 2030 and a net zero emissions by 2070. To achieve the targets the countries, have to go a great leap including India.

I. Objectives

The objective of the present paper is to:

- Analyse the various targets of carbon emissions set by various agreements;
- Highlight the current status of Green House Gas (GHG) emissions and target achievement by various countries;
- Sectoral level analysis of GHG emissions and relationship with India specific economic indicators

The study uses simple analytical techniques such as average, ratios, percentage methods etc. for analysis purpose.

II. Global Warming Potentials

Table 1 highlights the Global Warming Potentials (GWP) provided by the inter-government panel on climate change's sixth assessment report.

TABLE 1: SECTORAL WISE GLOBAL WARMING POTENTIALS (Co2EqGWP-AR6InMillions)

Sl. No.	Industries	2005	2018	% Increase
1.	Iron & Steel	105	246	134
2.	Non-metallic minerals	48	101	110
3.	Petroleum Refining	28	54	93
4.	Non-Ferrous Metals	22	81	268
5.	Chemicals & Fertilizers	18	75	317
6.	Pulp, Paper & Print	9	12	33
7.	Textiles & Leather	8.9	21	136
8.	Food Processing, Beverage & Tobacco	7.5	9.9	32
9.	Machinery	5.1	4.5	-12
10.	Other energy industries (Diesel & Oil extraction)	3.6	15	317
11.	Transport Equipment	1.5	4.8	220
12.	Wood & Wood Products	0.2	0.1	-50

Source: Council of Energy, Environment & Water

It may be noted that the GWP has increased for all industrial sectors except for Machinery and wood products during the period 2005 to 2018. The highest GWP increase was for Chemicals & Fertilizers and Other energy industries constituting about 317 percentage increase. This was followed by non-ferrous metals and transport equipment.

Table 2 shows the GHG emissions of selected global south countries in comparison to GDP and population.

	Total GHG emissions without LULUCFGgCo2Eqvt./ GDP (constant 2015 US\$ billion)		Total GHG emissions Without LULUCF GgCo2 Eqvt./Population	
	1994	2016	1994	2016
India	2191	1247	1.34	2.19
China (2014)	2458	1190	3.39	8.94
Brazil	600	582	3.92	4.95
Indonesia(2000)	950	1403	1.74	2.70
SouthAfrica	2027	-	9.83	-

<https://di.unfccc.int/indicators> non annex1#

The GHG emissions per GDP value has declined in countries including India except Indonesia during 1994 to 2016 period. India's GHG emissions per GDP value were 1247 GgCO₂ equivalent, which was higher than China and Brazil's emissions. The GHG emissions per population increased for all the given countries during the same period. But the acceleration of increase was much higher for China and stood at higher as 8.9 GgCO₂ equivalent per population, while it was 2.1 and 4.95 GgCO₂ equivalent for India and Brazil respectively.

III. Carbon trading

Stern and others addressed climate change as a market failure because of the greatest scale of distractions occurring in the world and stressed on widening the scope of enlarging the carbon markets (Stern et al., 2006). Carbon pricing is associated with Kyoto Protocol in 1997 with an objective to reduce the Green House Gas emissions (GHG). Carbon pricing has been defined by the World Bank as "*it is the cost of consuming carbon intensive fuels or using carbon intensive processes with social cost of doing so*". It is an economic incentive for the alterations needed in investment, production and consumption as well as to induce technological advancements. Carbon pricing can be direct as well as indirect. Direct carbon pricing includes carbon emission trading system (ETS), carbon tax and carbon credit mechanism. Under ETS, Govt. allows limit to Green House Gas (GHG) from the covered entities and the entities should surrender these emission units within the period and these emission units can be traded between covered entities. Under carbon tax, Govt. sets a price for the emissions. Carbon credits are issued to the processes that reduce emissions.

It is a financial incentive to reduce emissions which can be sold to generate revenue. The indirect carbon pricing is that which changes in the price of products that are associated with carbon emissions such as fuel taxes and subsidies (World Bank).

The Kyoto Protocol sets binding emission reduction targets of 37 industrialised countries. The first commitment owing to an average 5 percent reduction by 2008-12 as compared to 1990 GHG emissions. In the Doha amendment of 2012, a second commitment adoption was accepted by 147 parties for a period from 2013 to 2020. The United States had withdrawn from the Kyoto Protocol from 2001 citing economic damages (United Nations).

Carbon pricing mechanism includes direct and indirect pricing mechanisms. The direct carbon pricing includes emission trading system (ETS), carbon taxes, and carbon crediting mechanism via voluntary emission reduction strategies. The indirect carbon pricing includes the instruments that change the price of products associated with carbon emissions that are not directly proportional to relative emissions associated with product, eg. Fuel taxes (World Bank). The various instruments as implemented by various countries are given in the following table.

TABLE3: CARBON TAX & ETS COVERAGE IN THE WORLD

National Carbon Tax	Sub-national/state Carbon Tax	Regional Emission Trading Schemes	National ETS	Sub-national/state ETS
Finland, Poland, Norway, Sweden, Denmark, Latvia, Slovenia, Estonia, Switzerland, Iceland, Ireland, Japan, UK, France, Mexico, Portugal, South Africa, Chile, Colombia, Ukraine, Liechtenstein, Singapore, Canada, Netherlands, Argentina, Spain, Luxembourg, Uruguay, Hungary, Albania, Cote d'Ivoire, Senegal, Botswana, Indonesia, Israel, Morocco, New Zealand, Botswana, Kenya, Mauritania, Paraguay	British Columbia, carbon tax, Northwest Territories, Zacatecas, Taiwan, Queretaro, state of Mexico, Yucatan, Guanajuato, Durango, Catalonia, Manitoba, Jalisco, Hawaii, Manitoba, San Luis Potosi, Colima, Prince Edward Island, New Foundland, New Brunswick, Baja California, Tamaulipas, Alberta	EUETS, EUETS2, OBPS, RGGI	Switzerland, New Zealand, Kazakhstan, Korea, Australia, Canada, China, Mexico, Indonesia, Germany, Montenegro, UK, Austria, Chile, Colombia, Japan, Thailand, Turkey, Ukraine, Brazil, Vietnam, Malaysia, Pakistan, Gabon, Nigeria, India, Argentina	Alberta, Tokyo, California, Quebec, Shenzhen, Shanghai, Beijing, Guangdong, Tianjin, Hubei, Washington, Fujian, Newfoundland, Massachusetts, Nova Scotia, Saskatchewan, New Brunswick, Ontario, Oregon, Taiwan, China, Pennsylvania, Manitoba, Sakhalin, New York state, Colorado, Colima, Maryland, Ontario

Source: <https://carbonpricingdashboard.worldbank.org/about#download-data>

According to the world bank data, April 2025 about 36 countries adopted ETS, 39 countries with national carbon tax and 89 countries adopted sub-national carbon tax. The Regional emission trading system (ETS) includes European Union ETS, Output-Based Pricing system (OBPS) Canada and Regional Green House Gas Initiative (RGGI) in United States. India started the carbon tax on Coal in the year 2010 with a rate of Rs.50 per tone and later increased to Rs.400 per tonne. The European Union introduced the Emission Trading Scheme (ETS) in 2005. The EU ETS is one of the largest carbon markets in the world with a net worth of €770 billion in 2023. India is having an emission trading system (ETS) called Carbon Credit Trading Scheme (CCTS) implemented through Energy Conservation Act, 2001 and was notified in June 2023.

TABLE4: INSTRUMENTS OF CARBON CREDIT

Number	GHG emission coverage	Coverage of GHG emissions	Individual Percentage
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ETS	36	12.8 GtCO ₂ e	19 %ETS	31 % emissions from
Carbon Taxes	39	(24%)		highincome
	National&subnational89			countries
			6%carbontax	22%ofmiddle
				Income countries
Revenue	ETS	US\$75billion		
	Carbon Taxes	US\$29billion		

<https://carbonpricingdashboard.worldbank.org/compliance/coverage>

About 31 percentage of the emissions from high income countries and 22 percentage of the emissions from middle income countries are being covered under the above market instruments of ETS and carbon tax. Only 24 percent of the total global emissions are being covered by these instruments. Currently, ETS is generating revenue of US\$75 billion and carbon tax of about US\$29 billion. The price range is between US\$0.46 to US\$167 (World Bank). Table 5 explains the performance of major carbon credit markets.

TABLE 5: CARBON CREDIT MARKETS PERFORMANCE

COMPLIANCE MARKETS	2025	YTD (%)
European Union(EU)	€66.29	-11.84
UK	\$46.67	+31.03
Australia(AUD)	\$34.50	-4.83
New Zealand (NZD)	\$51.00	-18.40
South Korea	\$6.06	-8.01
China	¥75.76	-22.29
VOLUNTARY MARKETS		
Aviation Industry Offset	\$0.29	+123
Nature Based Offset	\$0.43	-25.86
Tech Based Offset	\$0.35	-

Source: <https://carboncredits.com/carbon-prices-today/>

There seems to be a downward price trend in the major carbon credit markets except for UK and aviation industry offset from January to April 2025. The larger downfall was for China, i.e. decline by 22 percent, followed by New Zealand and European Union, i.e. 18 and 11 percent respectively. The prices of Nature based offset, which is a voluntary market also witnessed 25.9 percent downfall trend during the same period.

Carbon Trading Exchange Platforms

Top carbon credit trading exchange platforms in the world includes Xpansiv New York, Aircarbon Exchange (ACX) Abu Dhabi, Carbon TradeXchange London, Intercontinental Exchange Dubai, Carbonplace Dubai, Toucan Dubai etc. Table 8 shows the major carbon exchange markets and its products.

TABLE 6: MAJOR CARBON EXCHANGE MARKETS IN THE WORLD

Sl. No.	EXCHANGES	INSTRUMENTS	EXPLANATION
1.	XPANSIV	Verified carbon standards	GHG crediting Programme

		ACR	Carbon Action Reserve Credits
		Cercarbono	Certified carbon standard
		Climate action Reserve	Voluntary Carbon market program with standards
		Gold Standard	Gold standard verified emission Reduction
		Puro Earth	Credit engineered carbon removal projects
2.	ACX		
		ACX Corsia Eligible tonnes	Aviation
		ACX global nature tonnes	Nature based project
		ACX global nature tonnes+ series	Nature based project & co-benefits
		ACX sustainable development goal tonnes	CC with SDG certification
		ACX Renewable energy tonnes	Renewable energy project
		ACX house hold off set tonnes	Improved cooking solution project
		ACX non-exchange carbon Tonnes	CC from Market registry which doesn't have any specifications
3.	Carbon Trade Xchange (CTX)		
		United Nations CERs	
		Gold Standard VERs	
		Verified Emission Reductions	
4.	Intercontinental Exchange(ICX)		
		California Carbon Offset Futures	Physically delivered California Air Resources Board greenhouse gas Emissions off set credits
		ICE Global Carbon Futures Index	
		ICEEUA Carbon Futures Index	It is a euro-denominated 1,000 Carbon Emission Allowances that are deliverable to or from the Union Registry under the European Union Emissions Trading System ("EU ETS"). It is an entitlement to emit one metric ton of carbon Dioxide equivalent gas.
		ICECCA Carbon Futures Index	
		ICERGGI Carbon Futures Index	
		ICEUKA Carbon Futures Index	
		ICEUS Carbon Futures Index	
5.	Toucan		
		CHAR Carbon Pool	Registry for high integrity carbon Credits

	BioCHAR CarbonPool	Soil carbon pools

Source: Author's compilation from various Exchange websites

TOKENISATION OF CARBON CREDITS

Tokenisation of carbon credits is a step ahead from traditional carbon credits towards digital token money in block chain. Reuters predict the estimated use of asset tokenisation adoption to be US\$16trn by 2028.

TABLE7: MAJOR EXCHANGES PROVIDING TOKENISATION OF CARBON CREDIT (2022)

Sl. No.	Exchanges	
1.	ToucanProtocol	- Toucan Protocol tokenised more than 20 million carbon credits
2.	ACX	- ACX tokenised
3.	MOSS	- 3.8million tCO ₂ e
4.	CarbonCredit (CCT)	- MOSS earth tokenised about 2.8 million carbon credits.
5.	ClimateImpactX (CIX)	
6.	BitmoPlatform	
7.	Flowcarbon	
8.	CarbonEco	
9.	JustCarbon	

Source: Price Water Coopers (PwC)

It may be noted that Toucan is a major exchange in Dubai done tokenisation of 20 million carbon credits. In future the tokenisation of carbon credits is expected progress.

IV. INDIAN CARBON MARKET MECHANISM

In order to contribute towards Paris Agreement, 2016 to limit the temperature rise by 2degree Celsius by the end of the century, India's commitment was to reduce the GHGs by 35-45 % by 2030 from 2005 levels. Indian industries were able to save 106 million tonnes of CO₂ emissions since 2015 to 2024. Further national framework for Indian Carbon Market (ICM) is being developed. ICM have two mechanisms, Compliance mechanism and offset mechanisms. Compliance mechanism addresses the emissions from the energy use and offset mechanism incentivizes voluntary actions from entities for GHG reductions. Specific carbon trading scheme provisions were incorporated under Energy conservation (Amendment) Act, 2022. This act enables the designated central agency to issue carbon credit certificates which represents one ton of CO₂ equivalent (tCO₂e) reduction from atmosphere.

The National Steering Committee for Indian Carbon Market under the Carbon Credit Trading Scheme (CCTS) recommended the Bureau of Energy Efficiency (BEE) to finalise the procedures for institutionalising Indian Carbon Market and to issue carbon credit certificate as well as to monitor the functions of Indian Carbon Market. The Bureau of Energy Efficiency is the administrative agency for the scheme, the Grid controller of India (GCI) functions as the registry operator for the scheme and manages Indian Carbon Market (ICM) registry and the Central Electrical Regulatory Commission (CERC) function as the regulator of trading activities under ICM. The validation and verification activities are to be conducted by Accredited Carbon Verification Agency (ACVA) (BEE, 2025). The Carbon Credit Trading Scheme (CCTS) has a compliance mechanism and offset mechanism. These mechanisms details are as given below: -

TABLE 8: CARBON MARKET MECHANISM IN INDIA																
	COMPLIANCE MECHANISM	OFFSET MECHANISM														
1.	Govt. sets GHG emission intensity targets for energy intensive industries	Voluntary Project based credit mechanism For the non-obliged entities for the issuance of Carbon Credit Certificates														
2.	BEE identifies sectors potential for reduction of GHG	Offset mechanism helps to tap mitigations from sectors not covered and incentivize actions.														
3.	BEE recommends the targets in terms of carbon dioxide equivalent(tCO2) per unit of equivalent product	It is possible for the non-obligatory sectors to register projections for GHG emission reduction and issuance of carbon credit Certificates upon fulfillment of eligibility.														
4.	Ministry of Power recommends the GHG emission intensity targets to MoEF under Environmental Protection Act,1986.	The project approval is based on detailed procedure for the offset mechanism before Issuance of CCC.														
5.	The obligatory entities that reduce their GHG emission intensity below the target shall be eligible for issuance of Carbon Credit Certificates & entities unable to achieve the target have to surrender/purchase equivalent number of certificates.	Sectors under offset mechanism <table><tr><th>Phase I</th><th>Phase II</th></tr><tr><td>-Energy</td><td>-Fugitive Emissions</td></tr><tr><td>-Industries</td><td>- Construction</td></tr><tr><td>-Agriculture</td><td>- Solvent use</td></tr><tr><td>-Wastehandling & Disposal</td><td>- Carbon Capture, storage of CO2</td></tr><tr><td>-Forestry</td><td></td></tr><tr><td>-Transport</td><td></td></tr></table>	Phase I	Phase II	-Energy	-Fugitive Emissions	-Industries	- Construction	-Agriculture	- Solvent use	-Wastehandling & Disposal	- Carbon Capture, storage of CO2	-Forestry		-Transport	
Phase I	Phase II															
-Energy	-Fugitive Emissions															
-Industries	- Construction															
-Agriculture	- Solvent use															
-Wastehandling & Disposal	- Carbon Capture, storage of CO2															
-Forestry																
-Transport																
6.	Perform, Achieve and Trade Scheme (PAT) will be transitioned to the compliance mechanisms under CCTS for wider decarbonisation.															
7.	Sectors such as Aluminium, Chlor Alkali, Cement, Fertilizer, Iron & Steel, Pulp & Paper, Petrochemicals, Petroleum refinery and textiles were considered for gradual transition and other Industries follows in near future.															

Source: <https://beeindia.gov.in/en/programmes/carbon-market>

Source: <https://beeindia.gov.in/en/programmes/carbon-market>

Offset mechanism is the voluntary project based credit mechanism for the non-obliged entities for the issuance of carbon credit certificate. Sectors under the offset mechanism in the first phase includes energy, industries, agriculture, Waste handling & Disposal, Forestry and Transport. The second phase sectors include Fugitive Emissions, Construction, Solvent use, Carbon Capture and storage of CO₂.

Table 9 shows the carbon neutrality programmes existing in India covering forest land, urban planning, air pollution, coastal ecosystem and regulations for climate resilience.

TABLE 9: CARBON NEUTRALITY PROGRAMMES IN INDIA		
Sl.No	Indicators	Programmes
1.	Forest Land Diversion and mitigation measures (Forest fragmentation is considering forest land for non-forestry purposes via	Compensatory Afforestation & National Afforestation Programme Ek Ped Maa Ka Naam tree plantation campaign launched on 2024 Green Credit Program to identify degraded forest land for tree plantation

	Van Adhiniyam, 1980)	
2.	Urban Climate Adaptation and Low-Carbon Development	- LT-LEDS emphasizing integrating adaptation measures & enhancing energy efficiency within the urban planning policies - Sustainable urban planning policies Include URDPFI, AMRUT, PMAY&SBM
3.	Air Pollution Control and clean air initiatives	- National Clean Air Programme for 131 cities - funding through various schemes such as AMRUT and fast adaptation of hybride and Electric vehicles.
4.	Coastal Ecosystem Conservation	- Mangrove & Coral Reef Conservation - Integrated coastal management plans - Mangrove initiatives for Shoreline Habitats & Tangible incomes (MISHTI) Programme
5.	Regulatory Measures for Climate Resilience	The coastal regulation zone (CRZ) notification 2011 & 2019 enhancing the climate resilience and targeting Management of mangroves, coralreefs, and other critical ecosystem.

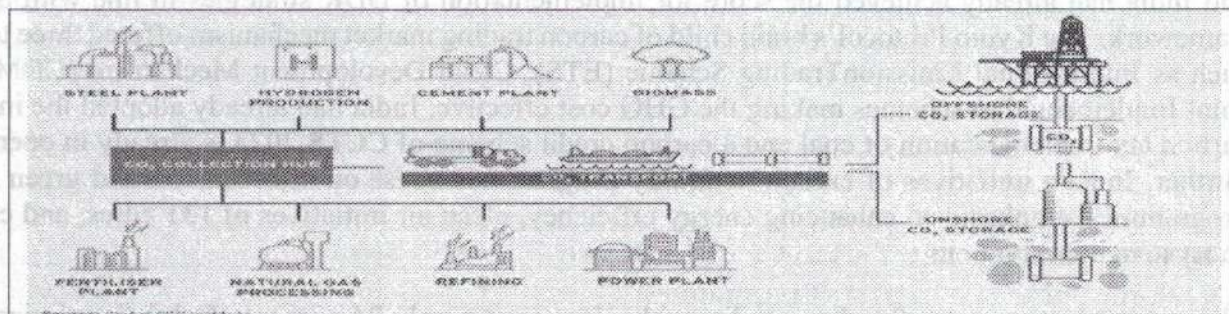
Source: <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2092311>

Some of the initiatives of Govt. of India's carbon neutrality programmes include compensatory afforestation and green credit programmes, urban climate adaptations with emphasis on enhancing energy efficiency, clean air initiatives of 131 cities, coastal ecosystem conservations by protecting mangroves and coastal regulation zone notifications for climate resilience.

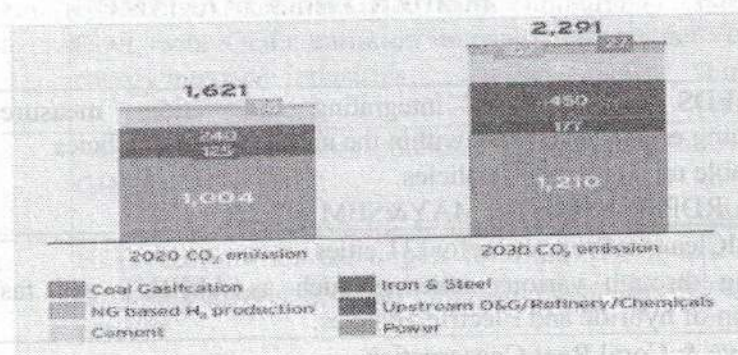
CARBON CAPTURE UTILIZATION AND STORAGE (CCUS) IN INDIA

According to Niti Aayog, the carbon credits based policy would establish a market for carbon based products and suitable for decarbonisation of existing industrial asset base. India as the 3rd largest emitter of CO₂ followed by US & China with an estimated emission of 2.6 giga tonne per annum. The carbon capture utilisation and storage plays an important role in decarbonising then industrial sector. The CCUS hub and framework is given in the below graph 2.

GRAPH 2: CCUS Hub and Framework



The carbon is being collected from various industries and biomass and is transported for offline and online storage. Industry wise CO₂ emissions in India are given in the following graph 3.

GRAPH3: INDUSTRY WISE CO₂ EMISSIONS IN INDIA (in mtpa units)

Power generation constitutes the maximum CO₂ emissions in India, i.e. about 1004 mtpa CO₂ equivalent in 2020, which is expected to increase by 1210 mtpa CO₂ equivalent because of over dependence on coal for power production. Other sectors are largely Iron & steel and Oil & chemical refineries.

India is expected to raise the CCUS funding via clean energy cess, Govt. Bonds and Budgetary support. Clean energy CESS on coal is expected to reintroduce from 2026 and expected to meet the short falls of funds by 2050 from all sources (Mukherjee & Chatterjee, 2022).

V. Summary & Conclusion

The public policy on climate change has created a market for carbon credit trades. Through carbon credit system the market failure issue due to negative externalities is being resolved, i.e. prices are being incorporated via purchase of carbon credits due to larger than permissible emissions. The product price may reflect this carbon price in future. Another issue is legalising the production of emissions by allowing companies to purchase carbon credits and rights for more emissions. The sale of permission of carbon credit certificates may enables the parties to produce more and emit more. The target of zero emissions is very long term i.e. 2050 and there are chances of extending it, for eg. zero emission targets for India is 2070. As per Keynes in long run we all are dead, so there is no idea whether policies will change in long term since the economic growth is always depend on produce more by exploiting more resources thereby creating a trade-off between environmental sustainability and growth. The Sustainable development goals on climate change majorly focus on adaptive capacities of countries, public policy implementation and educational awareness. The target achievement date for SDG is 2030 and India had already achieved the score for implementation of DDR strategies in line with Sendai framework. The Kyoto Protocol's brain child of carbon trading market mechanism offered three targets such as International Emission Trading Scheme (ETS), Clean Development Mechanism (CDM) and Joint Implementation schemes making the GHG cost effective. India had already adopted the indirect carbon tax implementation of coal and a carbon credit scheme of CCTS 2023 is already in operation. Further, India's initiatives of carbon neutrality programmes stress on afforestation and green credit programmes, emphasis on enhancing energy efficiency, clean air initiatives of 131 cities, and coastal ecosystem conservations.

The current instruments of carbon credit mechanism covers only 24 percent of global emissions is a serious concern. Moderate number of carbon credit markets, Exchanges and tokenisation of carbon credits are in operation but it is still in infant stage at global south. Industry wise CO₂ omissions are expected to increase in near future, i.e. 2291 mta units, so need need more decarbonisation efforts and more storage for carbon capture.

The market oriented carbon credit system to regulate carbon emissions has many constraints including the efficient market price of carbon credit, so a community involved Government regulated payment and controlling mechanism for the CO₂ emitters will be a more sustainable model to save the environment.

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